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SUPERSEDING
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MILITARY SPECIFICATION

TUBE, CENTRIFUGE, Polypropylene,
Graduated, Disposable

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of non-sterile, disposable, plastic centrifuge tube suitable for use in laboratory procedures.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATION

Federal

PPP-B-636	Box, Fiberboard.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6640

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U.S. DEPARTMENT OF COMMERCE

National Bureau of Standards (NBS)

NBS Circular 602 Testing of Glass Volumetric Apparatus

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office. Washington, DC 20402.)

3. REQUIREMENTS

* 3.1 Material. Tube shall be fabricated of translucent polypropylene; cap shall be fabricated of polyethylene or polypropylene.

* 3.2 Construction. Centrifuge tube shall consist of a tube with a screwcap in accordance with Figure 1. Surfaces of tube shall be smooth except for graduation markings, mold lines and gate stub. Wall thickness shall be uniform and shall be not less than .040 inch (in.) at the gate stub or in the area around the gate stub or in the conical portion at the bottom of the tube.

3.3 Graduation intervals. Tube shall be marked with lines at the following intervals: 2.5, 5.0, 7.5, 10, 15, 20, 25, 30, 35, 40, 45 and 50 milliliters (ml.). Each graduation line shall be marked with the appropriate number, and the letters "ml." shall follow the numbers.

3.4 Graduation markings. Graduation markings shall be raised, permanent and legible. Graduation lines shall be straight, shall be perpendicular to the vertical axis of the tube, and shall be a maximum of 0.6 millimeters (mm.) wide. Markings shall be on the interior or exterior surface of the tube.

3.5 Graduation accuracy. When tested as specified in 4.4.1, tube shall contain the amount indicated on the tube plus or minus the tolerances shown in Table I.

Table I Capacity tolerances

Graduation	Tolerances (ml.)
2.5 and 5.1 ml.	± 0.75
7.5 and 10 ml.	± 1.00
15, 20, 25 and 30 ml.	± 1.50
35, 40, 45 and 50 ml.	± 2.00

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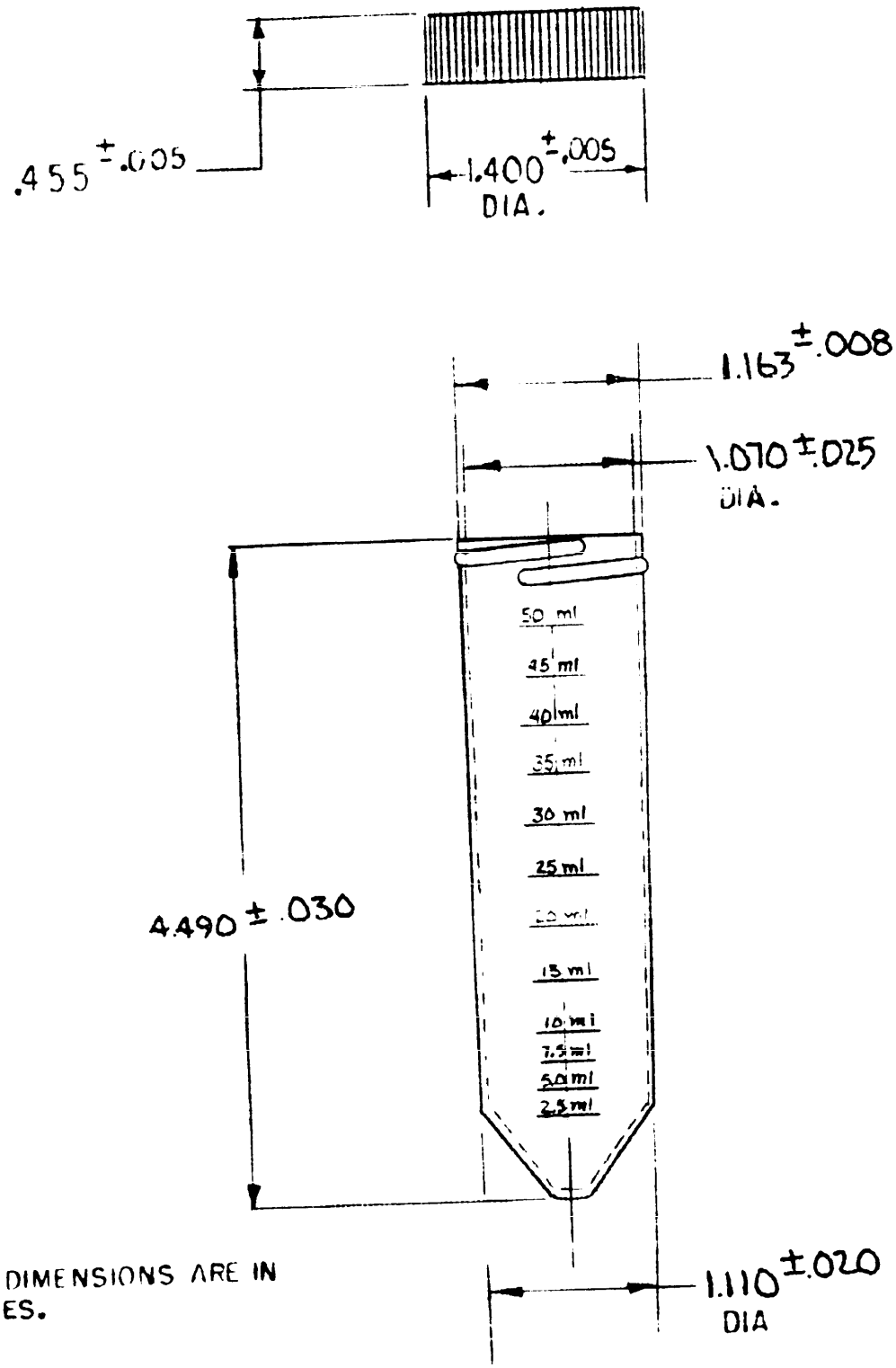


FIGURE 1

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3.6 Leakage. The screwcap shall make a watertight fit with the tube. Assembled centrifuge tube shall not leak when tested as specified in 4.4.2.

3.7 Identification marking. Centrifuge tube shall be legibly marked with the manufacturer's name or trademark.

3.8 Workmanship. Workmanship shall be first class throughout. Tube shall be free of defects listed in Table IV and other defects which detract from their appearance or may impair their serviceability. Gate stub at the bottom of cone shall not be considered a defect unless the stub is sharp or is longer than 1/2 in.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examination and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table II. Unit of product shall be one centrifuge tube (tube with screwcap).

Table II For examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

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* 4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table III. Unit of product shall be one centrifuge tube (tube with screwcap).

Table III For tests

Characteristic	Requirement	Test procedure	Inspection level	AQL
Accuracy	3.5	4.4.1	S-1	*
Leakage	3.6	4.4.2	S-1	.65

*Acceptance number shall be zero.

4.3 Examination. The centrifuge tube shall be examined to determine compliance with all requirements contained in this specification. Examination shall include but shall not be limited to the defects listed in Table IV.

Table IV Classification of defects

Category	Defects
Major	
101	Tube or screwcap not free of cracks.
102	Tube or screwcap not free of deformed areas.
103	Interior of tube not smooth (except for graduation markings).
104	Graduation markings not correct or legible.
Minor	
201	Tube or screwcap not free of dirt or other contamination.
202	Exterior of tube not smooth except for gate stub, mold lines and graduation markings and manufacturer's identification marking.
203	Identification marking not correct or legible.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Accuracy. The graduation markings of the tube shall be tested for accuracy in accordance with the test for "to contain" flasks and cylinders in N.B.S. Circular 602, paragraph 3.4. At least one graduation marking for each tolerance specified in Table I shall be tested.

4.4.2 Leakage. Tube shall be filled to the 50 ml. mark with a 50% alcohol solution. Screwcap shall be secured handtight to the tube. Centrifuge tube with contents shall be refrigerated for one hour at 10° Centigrade (°C.) ± 2°. Centrifuge tube shall be set on a flat surface with the cap facing downward and allowed to stand for one hour at ambient temperature. Appearance of liquid between the cap and tube shall be evidence of leaking.

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* 4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

* Table V Examination of preparation for delivery

Examination	Defects
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Levels A or C as specified (see 6.1).

5.1.1 Unit of issue. One package, containing 500 tubes with caps as specified, constitutes one unit of issue.

5.1.2 Level A.

* 5.1.2.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Box shall be closed as specified in the appendix of the box specification.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. No further packing is required.

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5.3 Marking.

5.3.1 Unit package. Each unit package shall be marked as required for the exterior container in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable level of packaging (see 5.1).

6.2 This specification covers the following item shown in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-00-054-7460	TUBE, CENTRIFUGE, Polypropylene, Conical Bottom, Graduated, 50 ml., Disposable. 500s

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
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